§ BUILD SPECIFICATION · RETAIL PARTNER DISPLAYS

Sunbox, Prism and Tower.

Displays that bring the sun indoors.

Fabrication brief for three consignment displays placed in hotel boutiques, airport retail and attractions. Each carries a UV-A ”sun” so guests can see Truecolors prints change colour indoors.

DOCUMENT	TC-DSP-001 Rev A
DATE	29 October 2026
ISSUED BY	True Colors Trading LLC, Dubai
CONTACT	hello@truecolors.global
UNITS	Millimetres unless stated
STATUS	For quotation and prototype. Not for production until shop drawings are approved.

§ 01 / OVERVIEW

Three displays, one job.

Truecolors tees are printed with UV-reactive ink: grey line art in the shade, full colour in direct sun. Indoor shop lighting has almost no UV, so every display carries its own sealed UV-A light that guests trigger with a button. The displays are placed on consignment in partner venues and must survive public use with little care from venue staff. Truecolors installs, restocks and services them.

	Sunbox	Prism	Tower
Code	TC-SBX	TC-PRS	TC-TWR
Type	Countertop, one demo shirt	Floor, 3-sided, display only	Floor, stock + design gallery
Overall (W × D × H)	420 × 300 × 600	600 per side (triangle) × 1850	1016 × 559 × 1829
Floor / counter area	0.13 m² counter	0.156 m²	0.57 m²
Demo shirts	1	2 (faces B, C)	20 designs (10 per side)
Screen	None (QR to video)	32" portrait, face A	None
Stock held	None	None, sold at till	~340 tees, 28 through-cubbies
Power	12 V plug-in adapter	Mains, 1 socket	Mains, 1 socket
Target mass	≈ 7 kg	≈ 80–90 kg (ballasted)	≈ 80 kg
Quantity, first run	Quote 10 / 25 / 50	Quote 3 / 10 / 25	Quote 2 / 5 / 10

Build the first unit of each type as a prototype for sign-off before batch production. The UV module (section 6) should be proven on a bench rig with real Truecolors shirts before cabinets are cut.

§ 02 / GENERAL REQUIREMENTS

Applies to every unit.

Brand finishes

Element	Specification
Carcass colour	Charcoal #232220. Matt 2K PU lacquer, 10–15 gloss units, or 0.8 mm matt HPL. Submit a drawdown for approval.
Interior of UV boxes	Off-white #FAF9F8, matt. No optical brighteners (they fluoresce under UV).
Gradient strip	4 mm band under every header: #C43FD3 → #E5603F → #F29A0C → #F2E600, printed vinyl or UV-print on acrylic. Artwork supplied.
Logo	Truecolors wordmark (white "true" + gradient "colors"), only ever on charcoal. Vector artwork supplied; never redraw.
Type on graphics	Space Grotesk (headings), JetBrains Mono (labels). All print-ready artwork supplied by Truecolors.
Edges & corners	2 mm ABS edge band or anodised aluminium angle (black). Exposed external corners radius ≥ 3 mm.

Materials

Part	Material	Notes
Carcass panels	18 mm moisture-resistant MDF (E1 / CARB P2), or 18 mm birch ply	Sunbox may use 15 mm. Ply for Prism faces if MDF chips at mitres.
Structure (Prism)	40 × 40 × 3 mm steel SHS spine, 10 mm steel base plate	Powder-coat black. Spine carries the screen VESA bracket.
Demo glazing	UV-filtering acrylic, ≥ 98% UV block below 380 nm	4 mm Sunbox, 5 mm floor units. e.g. ACRYLITE OP-3, PLEXIGLAS UF-3 or equivalent.

Part	Material	Notes
Shirt backboard	5 mm white PVC foam board	Shirt fixed with clear clips; quick to swap.
Header lightbox	5 mm opal acrylic, printed backlit film, 12 V LED module 4000 K	Even light, no hotspots at 300 mm viewing.
Graphic panels	3 mm aluminium composite (Dibond) or 3 mm white acrylic, UV-printed, matt laminate	Panels removable for re-branding per venue.
Fixings	Security Torx (pin) screws on public faces; knock-down cam fittings internally	All service doors on one keyed-alike cam lock (key code agreed with Truecolors).
Feet	M10 levelling feet, ±15 mm, nylon pads	Floor units only. Sunbox: 4 rubber feet + tether point.

Electrical & safety

→ All mains components certified for the country of installation (UL/ETL in the US, CE/UKCA in Europe and the UK, ECAS-registered in the UAE, NOM in Mexico). Ask Truecolors for the destination before ordering parts.

→ Mains enters through one panel-mount IEC C14 inlet with fuse and switch. Supply a 3 m cord with the local plug. Earth all exposed metal (Class I).

→ Everything inside except the screen runs on 12 V SELV from one enclosed PSU. Fuse each 12 V branch. Strain-relieve and tie every cable; no loose wiring in any shirt cavity.

→ Vent the electronics bays: slot vents top and bottom. Internal temperature must stay under 40 °C at 30 °C ambient (fit a thermostat fan if needed).

→ Floor units must not tip with a 50 N horizontal push at the top edge in any direction, or when tilted 10°. Provide floor-fixing points for airports.

→ UV light must never be visible directly from outside: no line of sight to an LED from any viewing position 0–1800 mm high. See section 6.

→ Fit a rating and serial plate on the rear: model code, serial, date, maker, voltage, power.

§ 03 / SUNBOX · TC-SBX

Sunbox, the countertop unit.

A small lit shadow box that sits beside the till. It holds one folded demo shirt behind UV-filtering glazing. Pressing the button lights the shirt for 20 seconds. It holds no stock; staff sell from behind the counter.



Design intent

Concept render for look and finish. Where it differs from the drawings, the drawings and this text govern.

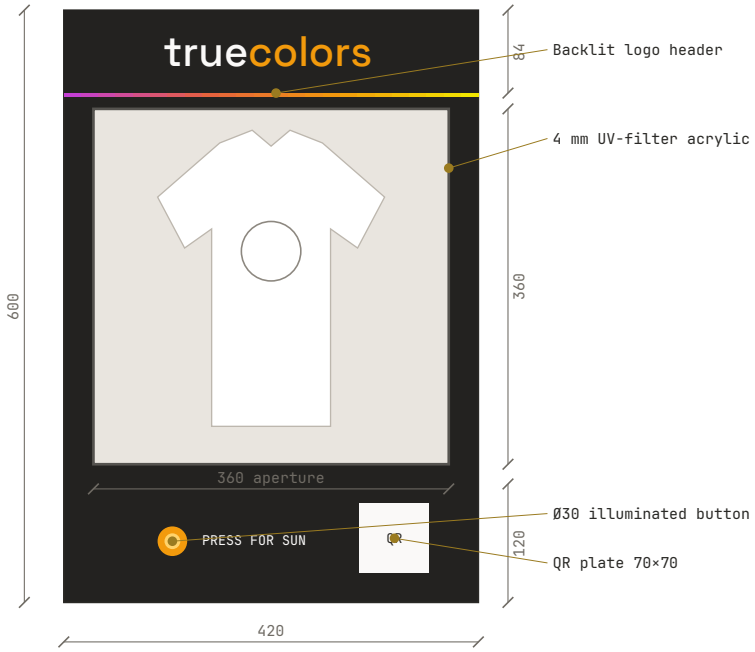
→ Reads as a lit picture frame: charcoal box, glowing header, bright window.

→ The violet glow is visible on the inside edges only; LEDs themselves are hidden.

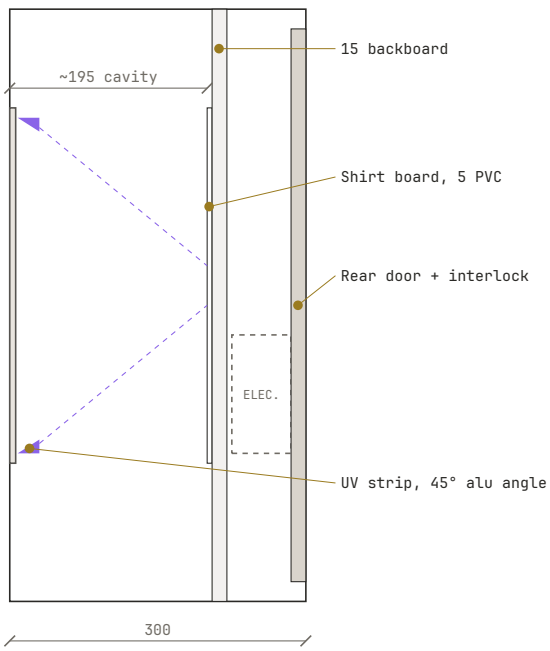
→ Button is the only colour on the base, so it's obvious what to press.

→ Same header, stripe and button as the floor units: one family.

FRONT ELEVATION



SIDE SECTION A–A



Construction

→ Five-sided 15 mm MDF box, charcoal outside, off-white inside. Front frame rebated 8 mm to take the 376 × 376 × 4 mm glazing, held by a screwed bezel (security Torx).

→ Header 420 × 84: lightbox with printed logo film over 5 mm opal acrylic; 4 mm gradient strip below.

→ Base zone 0–120: Ø30 illuminated button (left) and 70 × 70 QR plate (right) linking to the shade-to-sun video.

→ Shirt: adult tee retail-folded to about 250 × 330, clipped to a 5 mm PVC foam board 195 mm behind the glazing.

→ Rear lift-off door with cam lock and interlock switch: opening it switches the UV off. Electronics on the inside of the backboard.

→ Power: certified 12 V 2 A plug-in adapter, 5.5/2.1 mm DC jack on the rear, 2 m lead. Four rubber feet and a steel tether loop for a security cable.

Sunbox bill of materials (per unit)

Qty	Item	Specification	Est. USD
1	Cabinet kit	15 mm MR MDF, lacquered, per drawing	Fabricator
1	Glazing	376 × 376 × 4 mm UV-filtering acrylic	Fabricator
0.7 m	UV-A LED strip	12 V, 365 nm, 60 LED/m, in 2 × 330 mm runs	10–25
2	Aluminium angle profile	45° LED channel, no diffuser, 330 mm	4–8
1	Push button	Ø30 (or 22 mm), momentary, stainless anti-vandal, 12 V ring LED	8–20
1	Timer relay	12 V delay-off module, trigger input, set to 20 s	5–15
1	Interlock switch	Micro-switch, NC, roller lever	2–5
1	Header LED	12 V LED module / strip, 4000 K, ~2 W	3–8
1	PSU	12 V 2 A plug-in adapter, locally certified	8–15
1	Graphics	Header film, gradient strip, QR plate (artwork supplied)	Printer

§ 04 / PRISM · TC-PRS

Prism, the three-sided floor unit.

A triangular column that draws people in from across the floor. Face A plays a looping video on a 32” portrait screen. Faces B and C each hold one demo shirt behind UV glazing with its own button. The Prism holds no stock: graphics send guests to the counter for sizes.



Design intent

Concept render for look and finish. Where it differs from the drawings, the drawings and this text govern.

→ A slim triangular column: from any approach, guests see a screen or a lit shirt.

→ Sharp, trimmed vertical corners; flat triangular top cap.

→ Lower panels carry venue-specific line art in tone-on-tone grey (artwork supplied).

→ The lit window and the orange button are the brightest things on the unit.



Construction

→ Plan: equilateral triangle, 600 mm per face, 1850 mm high. Corners finished with black anodised 60° aluminium angle.

→ Structure: 10 mm steel triangular base plate (side 540) with a central 40 × 40 SHS spine, and 18 mm MDF triangular bulkheads at 100, 800, 1360 and 1646. Face panels screw to the bulkheads from inside.

→ Ballast: 2 × 8 kg steel plates bolted to the base plate. Three M10 levellers in a 25 mm recessed plinth with a 12 V LED floor wash. Three Ø9 holes for optional floor anchors.

→ Ships as two modules: body (0–1646) and crown (1646–1850), located with dowels and fixed with M8 bolts from inside. The crown has a plug-and-socket for the header LEDs.

→ Face A: screen on a VESA 200 × 200 bracket fixed to the spine; cut the aperture to the chosen screen's bezel +2 mm. The lower graphic panel (300–770) is the lockable access door to the electronics bay (PSU, strip, timers).

→ Faces B and C: the glazing frame (420 × 560 aperture, 5 mm UV-filter acrylic) is hinged on one side with a cam lock and interlock on the other. Shirt board sits ~80 mm behind the glazing on spine brackets. Use an adult

S/M tee with sleeves folded behind, or a youth L laid flat.

→ Buttons at 700 mm above floor (within accessible reach range). Screen centre at about 1240 mm (eye level).

→ Vent the screen bay at the top of face A and through the crown. Mass target 80–90 kg.

Prism bill of materials (per unit)

Qty	Item	Specification	Est. USD
1	Cabinet + crown	18 mm MR MDF, lacquered, aluminium corner angles	Fabricator
1	Steel frame	10 mm base plate, 40 × 40 SHS spine, 2 × 8 kg ballast, powder-coated	Fabricator
1	Display	32" commercial signage, portrait, ≥ 350 cd/m², 16/7 rated, USB autoplay, VESA 200 (e.g. Samsung QM32R, LG 32SM5J or equivalent)	450–900
2	Glazing	440 × 580 × 5 mm UV-filtering acrylic (in hinged frames)	Fabricator
1.8 m	UV-A LED strip	12 V, 365 nm, 60 LED/m: 2 × 420 mm per demo face	25–60
2	Push button	Ø30 momentary, stainless anti-vandal, 12 V ring LED	16–40
2	Timer relay	12 V delay-off module, 20 s	10–30
2	Interlock switch	Micro-switch, NC	4–10
1	PSU	12 V 60–100 W enclosed (Mean Well LRS series or equivalent)	20–35
1	Mains	IEC C14 fused inlet with switch, 4-way internal strip, 3 m local cord	20–40
3	Header LEDs	12 V LED modules, 4000 K, for 3 crown lightboxes	10–25
1	Fan (if needed)	80 mm 12 V quiet fan with 35 °C thermostat switch	10–20
1	Graphics set	3 header films, 3 gradient strips, 3 lower panels, QR (artwork supplied)	Printer

§ 05 / TOWER · TC-TWR

Tower, stock and a 20-design gallery.

A shop-in-shop for high-traffic boutiques, sized like the existing Magic Tees reference stand (40 × 22 × 72 in). Like the reference, it shows many designs at once: each side carries a gallery of 10 face-out tees behind UV glazing (20 designs in all), lit by its own button. The stock sits in 28 through-cubbies, open front and back, in four size columns with one design per row: as an island it is shopped from both sides, and against a wall it still works from the front, with no dead space behind.



Design intent

Concept render for look and finish. Where it differs from the drawings, the drawings and this text govern.

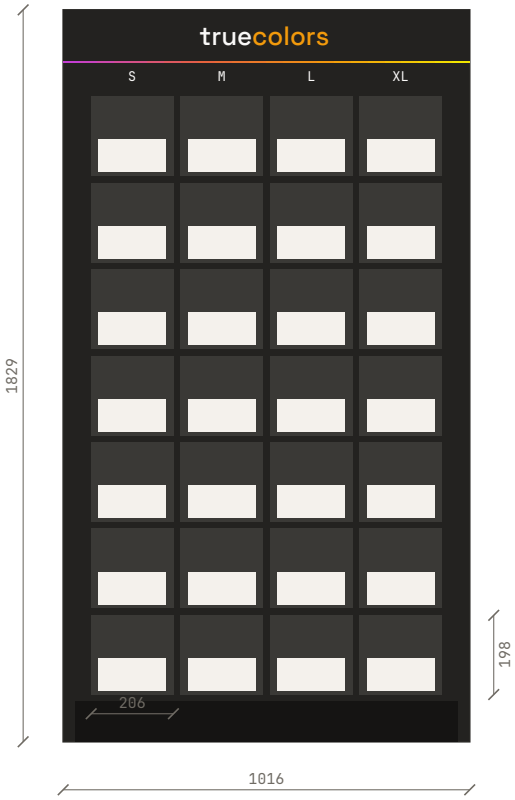
→ Galleries face the two approaches, so guests see lit designs from either side.

→ Front stock stays grey (no UV reaches it): the contrast with the lit gallery is the sales pitch.

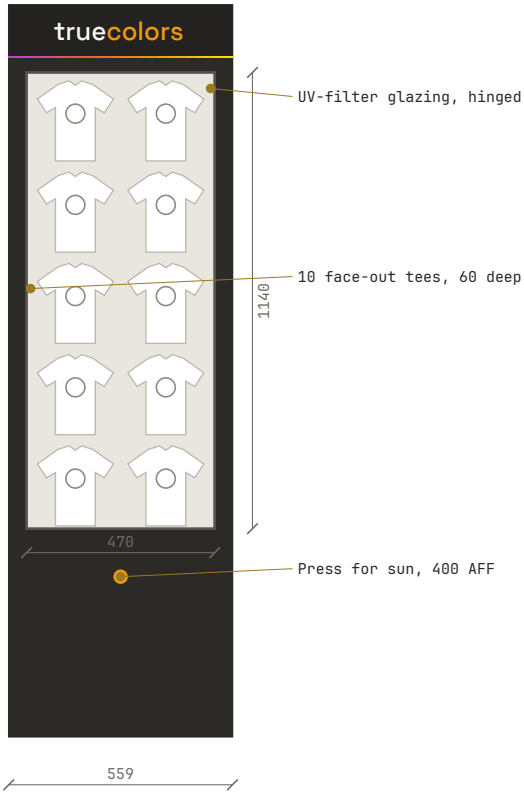
→ One design per row of cubbies, sizes S–XL across, as on the label rail; the back is identical.

→ Venue picks its 20 gallery designs; we swap them on restock.

FRONT = BACK ELEVATION — STOCK



SIDE ELEVATION — GALLERY (BOTH SIDES)



Construction

→ Carcass 18 mm MR MDF, 1016 × 559 × 1829, open front and back. Four columns (S, M, L, XL) by seven rows of through-cubbies, clear about 206 × 198 × 523, with a 22 mm high centre lip on each shelf so the front and back stacks stay separate; 70 mm side zones house the galleries.

→ Front and back faces are identical: header lightbox, S–XL label rail, 28 openings. Place as an island, or against a wall (restock from the front; the back openings simply face the wall).

→ Side galleries (both sides): aperture 470 × 1140 at 520–1660, 60 mm deep, off-white back. Ten folded tees face-out on 2 × 5 clear acrylic ledges (about 200 × 210 each).

→ Each gallery: 5 mm UV-filter acrylic in a hinged frame (piano hinge, cam lock, interlock), UV-A strips down both long edges, one Ø30 button at 400 mm above floor with its own timer.

→ Header lightbox wraps all four sides (129 high), gradient strip below. Size labels on a 40 mm rail above the cubbies, front and back.

→ No rear door: electronics (PSU, timers, mains strip) live in the header box behind a lockable lift-off top panel, with mains entering through the plinth.

→ Ships flat-packed: carcass, header and shelf kit with cam fittings. Four M10 levellers; anti-tip wall strap included.

Tower bill of materials (per unit)

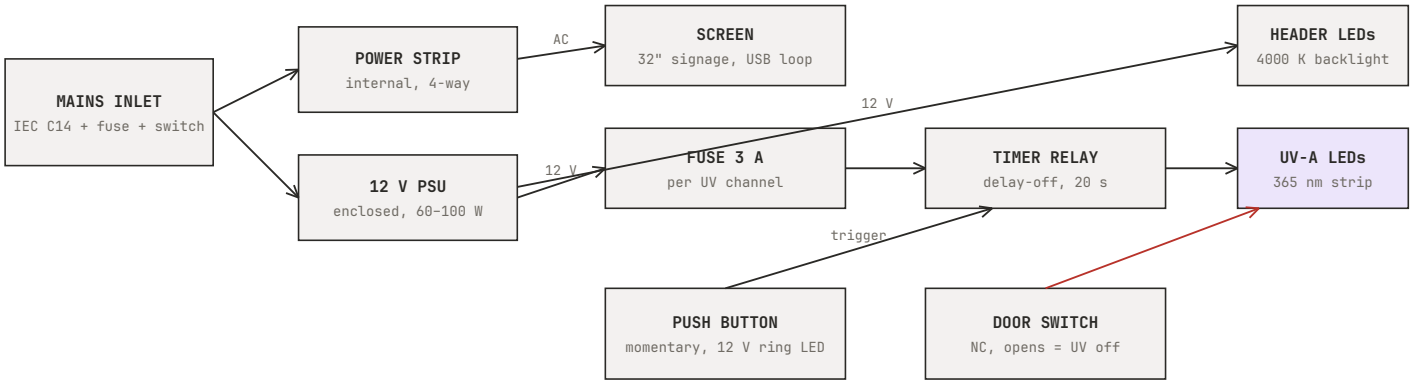
Qty	Item	Specification	Est. USD
1	Cabinet	18 mm MR MDF, lacquered, 28 through-cubbies with centre lips, two gallery bays, per drawing	Fabricator
2	Glazing	486 × 1156 × 5 mm UV-filtering acrylic, in hinged frames	Fabricator
20	Gallery ledges	Clear 3 mm acrylic ledge, 200 wide, with front lip	Fabricator
4.6 m	UV-A LED strip	12 V, 365 nm, 60 LED/m: 2 × 1140 mm per gallery (both long edges)	60–140
2	Push button, timer, interlock	As Sunbox, one set per gallery	30–80
1	PSU + mains	12 V 100 W enclosed PSU, IEC fused inlet, 3 m cord	40–70
1	Header LEDs	12 V, 4000 K	5–12
1	Graphics set	Four-sided header, gradient strip, size labels front and back	Printer

Est. USD covers bought-in components only, at small-batch prices; confirm locally. Cabinet, glazing, steel and print are quoted by the fabricator.

§ 06 / THE SUN MODULE

The UV demo module (all units).

This is the part that sells the shirt, so prove it first. Build a bench rig with one LED channel, one button and one timer, and test it on shirts Truecolors supplies before committing the cabinet design.



Sunbox: no mains inside. A certified 12 V 2 A plug-in adapter feeds the DC jack; the rest of the 12 V side is the same.

Prism has one timer + button + LED channel per demo face (2). The door switch breaks the LED supply, not the trigger.

Item	Requirement
Light source	UV-A LED strip, 365 nm peak, 12 V, 60 LEDs/m. 395–405 nm strips are cheaper but glow visibly violet and activate the ink less; use them only if the prototype test passes with them.
Placement	Two runs, along the top and bottom inside the glazing frame, in 45° aluminium angle aimed at the shirt. No PMMA/PC diffuser over the LEDs (it blocks UV).
Glazing	UV-filtering acrylic between guest and LEDs. The guest sees the colour change, not the UV.
Behaviour	Press: UV on for 20 s (adjustable 10–60 s), then off. Presses during a cycle are ignored. Button ring LED on whenever the unit is ready.
Interlock	NC micro-switch on every door or glazing frame that opens onto the cavity, wired in series with the LED supply. Door open = UV off.
Performance	Print visibly coloured within 10 s of pressing, full colour by 20 s, measured on a Truecolors demo shirt at 20–25 °C. Confirm on prototype.
Eye safety	Aim for the Exempt group under IEC 62471 at any viewing position. Fit a "UV-A inside: switch off before servicing" label inside every access door.
Option: press counter	A small microcontroller (e.g. Arduino Nano or ESP32) instead of the timer module, logging presses per day, so Truecolors can report engagement to the venue. Quote as an option.

Demo shirts lose some response after many cycles. Truecolors swaps them every 6–8 weeks during restocking; make the shirt board swappable in under 2 minutes.

§ 07 / SCREEN & GRAPHICS

Screen content and artwork.

Item	Specification
Screen behaviour	Powers on and plays the loop from USB automatically after a power cut, with no menus or input banners. Use the built-in scheduler to switch it off outside venue hours.
Video (supplied)	MP4, H.264, 1080 × 1920 portrait, 30 fps, 20–40 s seamless loop, no audio. Shows the shade-to-sun change, "Press for sun", and "Sizes at the counter".
Print artwork	Supplied by Truecolors as press-ready PDF, CMYK, 3 mm bleed, at 1:1. Maker supplies exact cut sizes from their shop drawings before artwork is finalised.
Venue panels	Lower graphic panels are swappable so each venue can carry its own location-specific design names.

§ 08 / ACCEPTANCE

Acceptance tests and handover.

#	Test	Pass criteria
1	Dimensions	Overall ±3 mm; apertures and joinery ±1 mm; glazing sits flush in its frame.
2	Finish	No visible defects at 1 m under shop lighting; colour matches approved drawdown.
3	Stability (floor units)	No tipping under 50 N at the top edge in any direction, or at 10° tilt, with screen fitted.
4	Electrical	Earth continuity and insulation test to local rules; 12 V branches fused; cables tied and relieved.
5	UV module	20 ± 2 s per press; re-press ignored mid-cycle; interlock cuts UV on every door; no LED visible from outside at 0–1800 mm.
6	Demo response	Truecolors shirt visibly coloured within 10 s of pressing.
7	Screen	Auto-plays the loop within 60 s of mains on; no menus; schedule set.
8	Burn-in	24 h powered with a button press every 5 min (automated); no faults, internal temperature under 40 °C.
9	Documentation	Photos of each unit, serial plate, wiring diagram inside the access door, 2 keys per unit, spare fuses.

Maker deliverables

→ Shop drawings and a cut list for approval before any material is cut.
→ Finish drawdowns (charcoal and off-white) and one sample of UV-filter acrylic.
→ Bench-tested UV module (section 6), then one prototype of each unit for sign-off.
→ Quotes at the quantities in section 1, with lead time, crating and delivery to the destination.
→ Ask Truecolors anything unclear before building, at hello@truecolors.global.